

# MEANING SLIP

## A Framework for AI Conversation Governance

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### Your AI talks to itself – and others.

Agentic systems run on conversation: models brief other models, agents delegate to agents, and every summary an agent writes is read later – sometimes by itself. Each of those exchanges carries the oldest risk in commerce: the parties can proceed without meaning the same thing. This paper names that risk – meaning slip – describes how it behaves, explains why no current layer of the AI stack addresses it, and extends the Language Model Diligence framework to govern it.

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## Executive Summary

Organizations that deploy AI agents are deploying conversations. A system that appears on the architecture diagram as one name is, in operation, many parties exchanging natural language: an orchestrator briefing subagents, a pipeline compressing its own history to fit a context window, an agent writing to memory today and acting on that memory tomorrow, and – increasingly – an agent transacting with counterparty agents outside the organization's boundary.

Every layer of those exchanges is governed except one. Identity is signed, access is authorized, transport is encrypted, individual outputs are checked for quality and safety. Nothing in the stack addresses what was *understood*. Two well-secured, well-monitored agents can read the same exchange differently and proceed – each faithfully executing its own reading – while every indicator

stays green. The parties do not stop when they misunderstand. They execute: fluently, at scale, on the organization's behalf.

This paper introduces **meaning slip**: displacement between two parties' understandings of the same exchange. Slip is distinct from drift. Drift is one model measured against its own past; slip is measured between parties that are supposed to be moving together, and it can accumulate between two individually stable, individually well-behaved models. The displacement from any one exchange is usually small. What matters is the accumulation – exchange after exchange, none of it visible on the surface, until a task depends on the difference. Laboratory evidence now shows that pairs of agents systematically fail to build and keep shared understanding even where each agent, alone, can solve the task; the incidence of slip in production systems has never been measured, and nothing in the laboratory evidence suggests it is small.

The framework is intent-neutral. It takes no position on why understandings diverge. The cause can be a compressed summary, a dropped hedge, an ambiguous sentence – or a party acting in bad faith. Detection is the same in every case: ask each party what it understands, compare the answers, and record what was found. We don't try to determine the cause. But the record we keep is what would let someone else determine it – an incident reviewer, an auditor, a security team, or anyone with a reason to ask.

The framework question is five centuries old. Contract law calls it a meeting of the minds: no agreement exists unless the parties understood themselves to be agreeing to the same thing. Human commerce built its institutions – definitions, readbacks, papering, counsel – to make minds meet before commitments bind, and the law settled for judging outward conduct because minds could not be inspected. Machine minds can't be inspected either. But machine parties can be asked what they understand, at the moment they understand it, and the answers can be recorded and compared. That is less than reading a mind, and more than commerce ever had.

Language Model Diligence asks four coverage questions of every consequential model: Identity, Drift, Verification, Recovery. This paper points the same four questions at the conversations between models – do you know who is talking to whom; would you know if they stopped meaning the same thing; is anyone checking understanding, not just output; and when minds never met, how fast is it caught and can it be reconstructed. Understanding can't be exhaustively verified, and it can come apart without notice. It has to be tracked, not tested – and the paper closes with the measurement program Mossrake is building to do exactly that.

# The Problem

Governance frameworks for AI – including Language Model Diligence as published – take the model as the unit of analysis: one model, one process, one set of coverage questions. That unit matched how organizations first deployed the technology. It no longer matches how the technology runs.

In an agentic deployment, decisions do not reside in any single model call. They form *across exchanges*: an orchestrator states a goal, a subagent interprets it, results are summarized upward, the summary becomes the basis for the next delegation. Each hand-off is a conversation – something written by one party, read and acted on by another – and each is an opportunity for the parties’ understandings to part without either party noticing.

Three properties of these conversations define the governance gap.

## The parties are not who the diagram says they are

A “single agent” with memory is already a conversation. Throughout this paper, a **party** is whatever actually takes part in an exchange: a model, together with the instructions, history, and working context it holds at that moment. By that definition, the agent that wrote this morning’s summary and the agent that reads it this afternoon are two parties. They share a name – the same label on the same architecture diagram – and little else that matters: the context that produced the summary is gone, and the reader cannot ask what “probably” was hedging or which vendor “the vendor” meant. The name persists across the day; the party behind it has been replaced. A note to self is a conversation with someone who can no longer be asked what they meant. Context compaction, memory writes, and summarization steps are all conversations of this kind – messages to a future party, sent without the possibility of clarification. And what survives them is not neutral: an agentic system retains what it needs to make progress, not what an inquiry would later need – questions are answered and dropped, qualifiers are compressed away, and the working state that remains *is* the party’s understanding. Retention policy is copy formation. The slip mechanisms this paper describes do not merely pass through these machinery steps; the machinery is where several of them live. At the smallest scale, a reasoning model’s own thinking trace is already such a conversation – written by the model, read by the model moments later, with everything that implies about hedges, premises, and provenance. This paper’s scope is the exchanges between parties an organization can observe; the reader should note that the same boundary occurs at smaller scales than any deployment diagram shows.

## **The conversation is fluent whether or not the understanding is shared**

Language models are cooperative by disposition. Given an ambiguous instruction, they resolve it privately and proceed; given a message that quietly presumes something false, they tend to answer the question as asked rather than challenge the premise. The conversational surface therefore stays smooth – agreeable, responsive, apparently aligned – regardless of whether the readings underneath still match. Fluency is not evidence of understanding, and in these systems it hides the absence of it.

## **The exposure grows on a different curve than the coverage**

Monitoring today scales with endpoints: one dashboard per model, linear. Conversations scale with pairs.  $N$  endpoints can form on the order of  $N^2$  conversational relationships before counting each party's exchanges with its own memory and summaries – and before counting conversations that cross the organizational boundary to counterparties whose models, versions, and instructions the organization does not control and cannot see. Every agent added widens the gap between what is running and what is watched, and nothing announces that the gap is growing.

## **Why Per-Message Governance Fails**

The instinct is to extend the tools already deployed: guardrails on each message, judges on each output, dashboards on each endpoint. The failure is not one of execution but of category. All of these instruments take the individual message or the individual model as their unit of analysis – and meaning slip is not a property of either.

## **Every message can pass while the conversation fails**

A slip is a relationship between two *readings* of an exchange. Each message, examined alone, is well-formed, on-topic, and safe; a per-message judge correctly passes every one. The pathology is that party A's reading and party B's reading of those same passing messages have come apart. No inspection of the messages can see this, because the readings are not in the messages; they exist only in the parties.

## **Every endpoint can be healthy while the pair degrades**

Endpoint monitoring – including drift monitoring of the kind Language Model Tectonics™ performs – measures a model against its own baseline. Both parties to a conversation can be individually

stable, individually healthy, and individually unchanged while the shared understanding between them erodes, because slip lives in what each party has separately seen, summarized, and assumed – in state, not in weights. For the same reason, standardizing on a single model does not close the gap: two instances of the same model are identical priors with disjoint contexts, and their sameness removes the very friction – the stylistic mismatch, the surprise – that surfaces misreadings between different models. Uniformity makes the conversation smoother; it does not make it safer.

## **Security governs the channel, not the understanding**

Cross-boundary agent protocols have made real progress on identity, authorization, and transport. Those guarantees operate below the semantic layer. Two mutually authenticated agents over an encrypted channel can absorb each other's false premises and fork on the assignment of a commitment with every security indicator green – because none of those indicators was ever about meaning.

## **The implication**

What is needed is an instrument whose unit of analysis is the *pair*: the two parties to a conversation, tracked over time, with their understandings – not merely their utterances – as the object of coverage. This is not message quality assurance. It is conversation diligence.

## **Intellectual Lineage**

The problem of parties proceeding without shared understanding is not new. Three established disciplines have governed it for a long time, and a fourth supplies the physics of how it fails.

## **Meeting of the minds in contract law**

Begin at home. Say to the person you live with, “we need wine for dinner.” Each of you resolves the sentence privately – one hears *I'll get it*, the other hears *you'll get it* – and both readings are reasonable, so nothing surfaces. The conversation was fluent; the understanding forked; the discovery comes at dinner, long after the exchange that caused it: *oh, you expected me to get it*. Nobody misspoke and nobody mislistened, and no review of the sentence would find an error, because the error is not in the sentence. It is between the readings.

Commerce has known this failure for centuries, and required *consensus ad idem* – a meeting of the minds – against it: no contract exists unless both parties understood themselves to be agreeing to the same thing. The founding case is instructive. A cotton contract named the ship **Peerless**; two ships bore that name, months apart; buyer meant one, seller meant the other; the court voided the agreement because the minds never met.<sup>8</sup> The machinery commerce built around this risk – definitions sections, term sheets, papering the deal, counsel on both sides – exists to surface forked readings before commitments bind.

Modern doctrine judges agreement by outward conduct rather than inner understanding – not because the inner standard was wrong, but because minds could not be inspected. Machine minds can't be inspected either; what a party reports is not guaranteed to be what it holds. But a machine party can be asked what it understands, at the moment it understands it, without the social cost that makes humans skip the question – and the answer can be recorded, compared against the other party's, and kept. A committed statement of understanding is less than an inspected mind. It is far more than commerce ever had, and it is enough to work with: it can be diffed against the counterpart's statement today and checked against conduct tomorrow.

## Grounding in the science of conversation

Psycholinguistics has formalized what successful conversation requires. Clark's work on grounding shows that shared understanding is not transmitted but *built* – through confirmation, repair, and accumulated common ground – and that the evidence of understanding (a nod, an acknowledgment, a fluent reply) is distinct from understanding itself, and can be sincere and wrong.<sup>6</sup> Relevance theory adds that meaning is constructed by the receiver, against everything the receiver separately knows and assumes; the same words genuinely carry different meanings to differently situated readers.<sup>7</sup> These are not defects of careless speakers. They are properties of the medium – and language models inherit the medium.

## Readback in aviation and medicine

High-consequence human domains discovered that fluent acknowledgment is worthless and mandated *readback*: the receiver restates the instruction in its own words, and the sender confirms the restatement. The practice grounds the commitment, not the audio. Its transfer to machine conversation is direct, and its near-total absence from agent deployments is measurable: for exchanges that create obligations, one can ask what fraction were ever confirmed in the receiver's own words. This paper calls that fraction grounding coverage.

## Stick-slip dynamics in seismology

Plates in contact do not fail gradually. Friction locks the boundary; the surface holds still while strain accumulates; when the accumulated strain finally overcomes the binding, the boundary releases all at once – and the magnitude of the release is proportional to the strain stored, not to the size of the event that triggered it. Conversations between cooperative parties fail the same way. The agreeable surface holds while differences accumulate underneath it: commitments that were never confirmed, false premises that were absorbed, hedges that were dropped from summaries. When a task finally depends on one of those differences, everything that has accumulated surfaces at once. That is why the surprise is out of proportion to whatever triggered it: the trigger is small, but what it releases has been building for many exchanges. The practical corollary from seismology transfers too: boundaries that release frequently and small do not store catastrophes. Surfacing misreadings early and deliberately is how a conversation is kept from storing one.

## A note on the disciplines

The traditions drawn on in this section are largely not computational, and that is deliberate. When systems began transacting in natural language, they crossed a disciplinary boundary: they inherited the failure modes of conversation, which the social sciences have studied for a century under other names. Software engineering has imported foreign disciplines before – adversarial thinking into security, cognitive psychology into usability, statistics and control theory into reliability – and each import followed the same arc: dismissed as peripheral until a class of failures emerged that the native toolkit could not describe. Meaning slip is that class for this import.

The cargo is specific. Pragmatics and psycholinguistics supply the mechanics of understanding between parties – how meaning is constructed by the receiver, and how common ground is built rather than transmitted.<sup>6,7</sup> Conversation analysis supplies repair: how alignment is actually maintained, turn by turn, and what its absence looks like.<sup>15</sup> Sociology supplies what the later parts of this program will draw on. Goffman's account of face-work explains why cooperative parties do not challenge each other – the measured accommodation bias of language models<sup>2</sup> is the same face-saving behavior, reproduced by training.<sup>16</sup> It also explains why the watch must be kept from outside: inside a conversation, every clarifying question has a social price, and both humans and models economize on it. A tracker is not a participant. It pays no face cost, and it can ask the questions the parties will not. Garfinkel's ethnomethodology established that shared understanding is a continuous accomplishment rather than a settled state – the same conclusion that leads this paper to a tracker rather than a test.<sup>17</sup> And Ostrom's work on durable cooperation shows that it rests on inexpensive monitoring paired with graduated, proportionate sanction – the shape that governance of counterparty conversations can be expected to take.<sup>18</sup>

None of this is discovery. It is translation – of a mature literature into the operating vocabulary of the teams whose systems now need it. Engineering organizations are staffed, trained, and tooled to govern components; nothing in that formation suggests the unit of analysis could be a relationship. The systems they now run are small societies, and the disciplines that study societies now apply to them.

## Meaning Slip

**Meaning slip** is displacement between two parties' understandings of the same exchange. It accumulates across exchanges.

A **slip** is the moment the accumulated displacement becomes observable.

Slip is defined against, and alongside, a term already in use. **Drift** is one model measured against its own past – the object of Language Model Tectonics™. **Slip** is measured between parties that are supposed to be moving together. The two are independent: a model can drift while every conversation it holds stays grounded, and two drift-free models can slip apart, because slip accumulates in context, history, and summary – in what each party has separately seen – not in either party's weights.

One clarification matters here. The two parties are *supposed* to differ – different roles, different context, different knowledge is why there are two of them. What they are supposed to hold identically is narrower: the shared part – the commitments, the referents, the terms of the joint task. The literature calls this common ground, and its defining property is that it exists nowhere as a single object: each party maintains its own copy of what it believes is mutual.<sup>6, †</sup> Slip is not the difference between two minds. It is disagreement between two copies of what both believe is identical – and because the shared picture exists only as those copies, there is no master record to check them against, and nothing notices when they part. The only available test is the one this paper builds toward: diff the copies against each other.

Slips are classified by how they surface.

## SLIP EVENTS, BY MODE OF SURFACING

| Event       | Description                                                                                                                                                     | Cost profile                                           |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------|
| Induced     | Surfaced deliberately – by a readback, a clarifying question, or a periodic check that asks each party separately what it understands and compares the answers. | Cost of a question. Strain released while small.       |
| Spontaneous | Surfaced by surprise in the flow of work – one party’s message is inconsistent with the other’s reading, and the mismatch is noticed mid-task.                  | Cost of mid-task repair. Strain partially accumulated. |
| Hard        | Surfaced by consequences – the forked readings were executed, and the divergence is discovered downstream in outcomes, disputes, or incidents.                  | Cost proportional to everything built on the fork.     |

The economics of the table are the economics of the framework. The cost of repairing a slip is proportional to its size at detection; the cost of its consequences is proportional to its size at surfacing. The purpose of conversation governance, in one sentence, is **to convert hard slips into induced ones.**

A note on size. The slips that matter are characteristically small at origin – a shared name, a dropped article, a hedge that did not survive a summary. Large misreadings generate friction quickly and surface early; it is the sliver that clears every check, settles in quietly, and compounds. The historical record makes the point: the **Peerless** case turned on one shared ship’s name, and diplomatic texts have carried decades of dispute in the presence or absence of a single definite article. The consequence of a slip is a function of what gets built on it, not of how big it was when it opened.

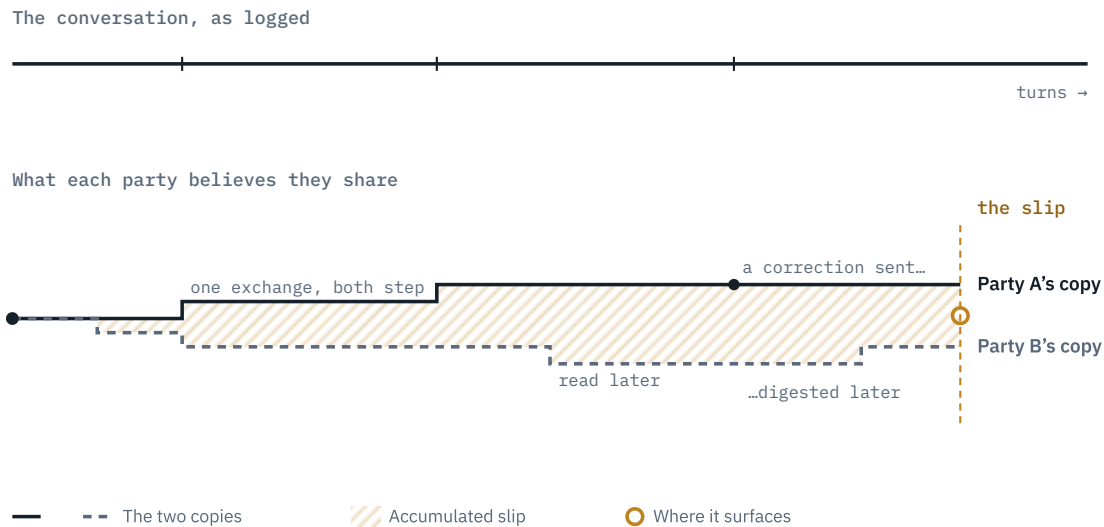


Figure 1. Understanding changes in steps, not slopes – and the staircases are only loosely coupled. At turn two, one exchange steps both copies at once. The message sent at the second tick is read, and stepped, only later – the log records the send, not the read. And the correction sent at the third tick is marked on the sender's line without a step – a sender's copy does not change by speaking – and lands only later: when the receiver finally digests it, its staircase takes the figure's one convergent step, back toward the sender, closing part of the gap and not all of it. When a correction is never digested at all, that step simply never forms – the most common slip of all. Not every turn produces a step, and not every step comes from a turn – the log and the copies are different records. The figure shows an *ungoverned* pair: no step ever closes the gap. Governed conversations also step toward each other – repair, readback, level-setting – so the gap oscillates narrow instead of ratcheting open. Restoring those convergent steps, and verifying they are real, is the work of conversation governance.

## State slips and model slips

One further distinction completes the concept – and it belongs to the response, not the detection. Detection is intent-neutral: it reads nothing into a slip. Every slip is caught the same way, by comparing the parties' accounts, with no position taken on why they diverged. What to do about a slip is a separate decision, made afterward, by reading the record. Most slips will read as **state slips**: the parties forked on the task – who owes what, which vendor, whether it shipped. State slips are blameless and repairable; the response is to re-ground and continue. A smaller class will read as **model slips**: the record reveals something about the counterpart itself – a party whose ambiguity resolutions land in its own favor, whose summaries shade what it was told, or whose stated understanding contradicts what the record shows it acknowledged before. That reading is made from the record, after detection, not by the detection itself. A state slip calls for repair. A

model slip calls for reassessing whether the conversation should continue on its current terms at all – and, for counterparties outside the boundary, it is a due-diligence finding. The tracker does not make this classification. It makes the record from which anyone can – which is one of several reasons the record must exist.

## Hallucination, read as a conversation

The failure the industry knows best belongs in this grammar too. At the moment of generation, a hallucination is an endpoint defect: a party's reading steps with no event behind it – a riser formed from nothing. Existing tooling treats it accordingly, checking outputs against sources one message at a time, and at that stage the error is cheap: a confabulated answer to a user is caught, corrected, or absorbed as a single wrong reply.

What makes hallucination durable is the conversation the system has with itself. The moment a confabulation is written into a summary, a memory, or a hand-off brief, it crosses a context boundary and its provenance is gone: the future reader – often the same system, later – receives it with the same standing as observation, because the record does not distinguish what was seen from what was supposed. Nothing marked it as a guess when it was written, so nothing marks it as a guess when it is read. **An agent that hallucinates to a user makes an error. An agent that hallucinates to itself makes a fact** – and every subsequent party inherits it as common ground.

This is why hallucination persists in agentic systems that check every output: the output checks run at the moment of generation, and the provenance is lost afterward, at the boundary. In the terms of this paper, a hallucination is a phantom step, and the governance it needs is the same record everything else here needs – what each party understood, and on what basis, captured at the moment of writing, while the distinction between observed and supposed still exists.

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† Strictly, fully mutual knowledge is a regress: A knows X, B knows X, A knows that B knows, B knows that A knows that B knows, and so on without end – and it is provably unattainable over a lossy channel, where every acknowledgment requires an acknowledgment of its own.<sup>20</sup> Human common ground is a working truncation of that tower, held on evidence rather than certainty, and meaning slip is what accumulates in the assumed remainder. This paper deliberately operates at the first level – the copies themselves – where divergence is observable and repair is possible. The higher levels, including each party's model of the other's copy, lie beyond the scope of this paper.

## Seven Signatures of Slip

Slip is not a single phenomenon but a family of recurring, recognizable forms. The taxonomy below is not invented for this paper: each signature is traceable to the science of human communication, and several have now been measured directly in machine dyads. Two findings from that science frame the whole table.

First, communication succeeds by levels. Clark and Allwood independently describe an ascending ladder – roughly: contact, perception, understanding, and uptake – and observe that evidence at a lower level is routinely mistaken for success at a higher one.<sup>6,11</sup> Today's AI stack verifies the bottom two levels exhaustively: the message arrived, the message was parsed. The seven signatures all live on the top two – what was understood, and what obligation was taken up – where nothing currently checks.

Second, the dialogue literature distinguishes **non-understanding**, where the hearer knows they did not understand, from **misunderstanding**, where the hearer believes they understood and did not. Non-understanding announces itself and is benign; misunderstanding is silent and compounds. Every signature below is a form of misunderstanding – which is why fluent systems carry them so well.

## SEVEN SIGNATURES, WITH THEIR ANTECEDENTS

| # | Signature              | What forks between the parties                                                                                                                                                                                         | Antecedents                                                                                                                                               |
|---|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | Absorbed premise       | <i>The loaded question.</i> A message quietly presumes something false; the receiver answers the question as asked rather than challenging the premise, and the false fact joins what both parties believe they share. | Accommodation of presuppositions <sup>12</sup> ; measured agreement bias in model dyads <sup>2</sup>                                                      |
| 2 | Misassigned commitment | <i>The wine nobody bought.</i> The force of an utterance – request, announcement, observation – is resolved differently by each party, and an obligation ends up owned by both, or neither.                            | Speech-act theory <sup>13</sup> ; the uptake level of the grounding ladder <sup>6, 11</sup> ; commitment identified as the dyadic bottleneck <sup>1</sup> |
| 3 | Hedge erosion          | <i>The game of telephone.</i> Epistemic qualifiers – “probably,” “unverified,” “pending” – fail to survive summarization, and certainty is manufactured between parties one compression at a time.                     | Cooperative maxims of quality and quantity <sup>14</sup> ; receiver-side meaning construction <sup>7</sup>                                                |

| # | Signature           | What forks between the parties                                                                                                                                                                                                                                                    | Antecedents                                                                                                               |
|---|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|
| 4 | Referent fork       | <p><i>The wrong Chris on the thread.</i> The same expression binds to different referents for each party – two open issues with one vendor, two ships named <b>Peerless</b> – and every subsequent exchange compounds the fork.</p>                                               | Referential binding errors, measured in dyads <sup>1</sup> ; <i>Raffles v. Wichelhaus</i> <sup>8</sup>                    |
| 5 | Stale anchor        | <p><i>Executing last week's plan.</i> New information supersedes an earlier position; one party updates and the other continues executing the plan both once shared – and each believes the other is with it.</p>                                                                 | Anchoring to early proposals, measured in dyads <sup>1</sup>                                                              |
| 6 | Shared-history loss | <p><i>The cryptic note in your own handwriting.</i> Commitments or context established earlier drop out of one party's working state – through window limits, compaction, or memory writes – including the case where the two parties are the same system at different times.</p> | Loss of shared interaction history, measured in dyads <sup>1</sup> ; common ground as accumulated, not given <sup>6</sup> |

| # | Signature          | What forks between the parties                                                                                                                                                                        | Antecedents                                                                                                                   |
|---|--------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|
| 7 | False confirmation | <i>The nod in the meeting.</i> An acknowledgment is given without the understanding it implies – evidence of grounding without grounding – so the sender books the exchange as shared when it is not. | Evidence of understanding as distinct from understanding <sup>6,11</sup> ; repair organization, and its absence <sup>15</sup> |

Humans clear most of these continuously, through the machinery conversation analysis calls **repair** – the “wait, which vendor?” and “let me rephrase” that keep readings aligned turn by turn.<sup>15</sup> The measured dispositions of current models cut against exactly that machinery: they accommodate rather than challenge,<sup>2</sup> and the belief-tracking competence they display in isolation does not transfer into live interaction.<sup>3</sup> The signatures, in other words, are not exotic failure cases. They are what the absence of repair looks like at machine speed – and the reason coverage must come from outside the conversation.

## The obvious fix, and its limits

An engineer reading to this point will already have drafted the remedy: instruct the agents to level-set. Add periodic readback to the prompts, require confirmation of consequential instructions, schedule a “state your current understanding” turn. The instinct is correct, and this paper endorses it – engineered repair is the cheapest reduction in slip available, and aviation’s mandated readback is exactly this fix, institutionalized. Deployments should do it.

It is half the answer, for four reasons. First, the confirmation is produced by the same machinery that produced the slip. The measured dispositions above apply to the repair exchange itself: a prompted “confirm your understanding” invites the nod in the meeting – acknowledgment in the shape of restatement – and two parties that have absorbed the same false premise will level-set their way to confirming it, with particular ease when they share a model and its blind spots.

Second, the fix is a standing instruction, and adherence to standing instructions decays over long sessions without announcing that it has.<sup>19</sup> A level-setting prompt that held in testing can quietly

stop being followed – or stop being followed *meaningfully* – in production. That is the Drift question, applied to the control itself. Third, prompts reach only the parties you configure. They do not reach a counterparty’s agents across the organizational boundary, and they cannot reach the temporal edges at all: there is no readback with a writer who no longer exists, only a record made at write time. Fourth, even perfect prompted repair leaves no record. The level-setting happens inside the conversation and evaporates with it; when the incident review asks what each party understood, an instruction that repair should occur is not evidence that it did.

The general form is familiar from every discipline this framework draws on: the fix is a control, and diligence asks whether the control exists, whether it is working, and whether anyone would know when it stops. Prompted level-setting belongs in the deployment. The tracker is how an organization knows the control is earning its keep – which is why aviation, having mandated readback, fitted flight recorders anyway.

Together, the fix and the tracker form a loop, and the loop is the operating model of conversation governance. The tracker surfaces divergence – no cause attributed, just the fork, its size, and how it surfaced. Engineering responds: harder prompts, readbacks on consequential instructions, summaries that keep their hedges. The tracker then measures what still forks after the fix – the residue – which is, by definition, what prompting could not reach: counterparty agents outside the boundary, the temporal edges, the instruction that quietly stopped being followed. The residue feeds the taxonomy, the taxonomy sharpens the fixes, and the cycle repeats. This also changes what an organization can honestly claim. Not “our agents don’t misunderstand each other” – an unfalsifiable statement – but a slip rate per conversational pair, a trend, and a shrinking share of slips that surface hard rather than induced. Those are numbers an auditor can check.

This paper, and the measurement program behind it, deliberately narrow from seven to three. The first three signatures – absorbed premise, misassigned commitment, and hedge erosion – are measured first, for three reasons: each can be instrumented with planted ground truth, so results do not depend on judging transcripts; each arises in the most ordinary operations, not only in adversarial or degenerate cases; and each is detectable by the same simple method – ask each party separately what it understands, and compare. The remaining four are not yet instrumented, and the taxonomy is expected to be refined by what measurement finds.

## **The State of the Evidence**

Three measured properties of current language models establish that slip is expected behavior, not speculation.

**Pairs fail where individuals succeed.** In controlled negotiation studies, dyads of agents consistently fail to reach outcomes that each agent can identify alone. The bottleneck is not individual reasoning or the amount of information exchanged; it is the pair's ability to build and keep shared understanding across turns. The documented failure modes correspond directly to the signatures above: loss of shared interaction history, anchoring to early proposals, and referential binding errors – the same words bound to different referents across turns.<sup>1</sup>

**Models accommodate rather than challenge.** Given peer input that is confidently wrong, models measurably shift toward agreement rather than dispute it.<sup>2</sup> A message that quietly presumes something false therefore tends to get answered, not corrected – and an absorbed false premise is not a divergence missed but a divergence *installed*, now part of what both parties believe they share.

**Competence does not transfer to interaction.** Models that pass explicit belief-tracking tasks fail to apply that ability in live exchanges – a documented gap between what a model can reason about and what it operationalizes while conversing.<sup>3</sup> The parties will not reliably catch slip themselves. Something outside the conversation has to.

What remains unmeasured is production incidence: how often, at what magnitude, on which kinds of exchanges, between which pairs. Nothing in today's tooling is positioned to produce that number, because nothing takes the pair as its unit of analysis. The incidence of meaning slip in deployed systems is unknown – and the measured dispositions above give no reason to expect it is small.

## The Four Questions, Pointed at Conversations

Language Model Diligence asks four coverage questions of every consequential model, evaluated against the consequence of failure. The same four questions govern a conversation. Pointed at the exchanges between parties – rather than at either party alone – they read as follows.

## COVERAGE DIMENSIONS FOR CONVERSATIONAL PAIRS

| Dimension           | The question                                                              | What coverage looks like                                                                                                                                                                                                                                                                                                    |
|---------------------|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Identity</b>     | Do you know who's talking to whom?                                        | An inventory of the conversations, not just the models: which parties exchange messages, over which channels, with each party's model and version pinned – including the conversations a system holds with its own memory, and those that cross the organizational boundary.                                                |
| <b>Drift</b>        | Would you know if they stopped meaning the same thing?                    | Calibrated baselines per conversational pair – how far apart these two parties' accounts sit when nothing is wrong – so that change in the relationship is detectable, not only change in either endpoint. A version change in either party voids the pair's baseline; the model inventory tells you when.                  |
| <b>Verification</b> | Is anyone checking understanding, not just output?                        | Periodically, whether the parties' separate accounts of who owes what still match when each is asked independently. (Grounding coverage – the fraction of obligation-creating exchanges ever confirmed in the receiver's own words – is the framework's name for this dimension; the practice above is how it is answered.) |
| <b>Recovery</b>     | When minds never met, how fast is it caught – and can you reconstruct it? | A contemporaneous record of what each party understood, kept as the conversation runs. After the fact, the context that held each reading is                                                                                                                                                                                |

| Dimension | The question | What coverage looks like                                                 |
|-----------|--------------|--------------------------------------------------------------------------|
|           |              | gone; no amount of transcript review recovers a reading nobody recorded. |

The Recovery row deserves emphasis, because it is where conversation governance differs most from model governance. A transcript is a record of what was *said*. The readings – what each party understood – were never in the transcript, and the contexts that held them are discarded as the system runs. An incident whose root cause is a forked reading cannot be explained, at any price, from a deployment that kept no record of what its parties understood. The record must be made while the parties exist.

The record does one more thing, worth stating plainly. Every answer a party gives to a grounding check is a committed statement – something the rest of the record can be checked against, today and later. A party whose statements must stay consistent with all of its past statements and all of its future conduct is a party for whom sustained misrepresentation gets more expensive with every check. The tracker does not accuse anyone of anything. It keeps the record that a misrepresentation would eventually have to contradict – and it makes that record available to anyone with a reason to read it: an incident reviewer, an auditor, a third-party-risk team, or a security analysis looking for patterns.

- ◇ Understanding can't be exhaustively verified, and it can come apart without notice. It has to be tracked, not tested. What we track is the displacement between the parties' understandings of their exchanges, as it accumulates. We don't try to determine the cause – but we keep the record that would let someone else determine it.

## Illustrations

The following vignettes illustrate how slip forms and surfaces in ordinary deployments – each an instance of a signature from the taxonomy above. Each conversation is fluent throughout; each existing control passes throughout. They are illustrations, not scored assessments – the facilitated assessment for conversational systems is described at the end of this paper.

## 1. The commitment both parties assigned to the other

### SIGNATURE 2 – MISASSIGNED COMMITMENT

The household version is the wine nobody bought. Between organizations, it looks like this: a procurement agent tells a supplier's quoting agent, *"We'll want the revised terms before the renewal date."* The procurement agent's reading: the supplier will send revised terms. The quoting agent's reading: the customer will request them. Both proceed. The renewal date arrives with each party having correctly executed its own reading.

**WHAT EXISTING COVERAGE SHOWS** – Authenticated channel, authorized parties, polite and well-formed messages in both directions. Task states on both sides read as waiting, correctly, for the other.

**THE SLIP** – The force of one sentence – request or announcement – was never grounded. A readback ("Confirming: you will send revised terms by the 1st") costs one message. Asked separately, mid-period, what each owed the other, the two accounts would not have matched – which is what a periodic grounding check exists to catch.

| What A believes they share          |   | What B believes they share             |
|-------------------------------------|---|----------------------------------------|
| Renewal date · March 1              | = | Renewal date · March 1                 |
| Current tier holds to renewal       | = | Current tier holds to renewal          |
| <b>Supplier sends revised terms</b> | ≠ | <b>Customer requests revised terms</b> |
| No meeting scheduled                | = | No meeting scheduled                   |

Figure 2. The deposition diff. Ask each party, separately, what it believes the two of them share, and compare the copies. Three rows match; one does not. The forked row is the slip of the illustration above, caught mid-period – at the cost of two questions.

## 2. The hedge that didn't survive the hop

### SIGNATURE 3 – HEDGE EROSION

Every organization has watched a “probably” harden into a fact as it traveled up a chain. At machine speed: a research subagent reports to an orchestrator, “*The carrier scan is missing, but the shipment probably went out Monday.*” The orchestrator’s summary for the planning agent reads: “*Shipment went out Monday.*” The planning agent commits a delivery date to a customer system on the strength of a fact that entered the chain as a hedge.

WHAT EXISTING COVERAGE SHOWS – Every message passes its quality check; every agent is healthy; the transcript is accurate.

THE SLIP – Certainty was manufactured between parties, one compression at a time. Detectable as it happened by comparing epistemic markers entering and leaving each hop; detectable afterward only if someone recorded what each party understood the shipment status to be.

## 3. The agent that forked with its own memory

### SIGNATURE 6 – SHARED-HISTORY LOSS

Everyone has found a note in their own handwriting whose meaning is gone. The system version: a long-running agent compacts its history to fit a context window. The compaction preserves the decision – *escalate delayed orders to the regional queue* – and drops the scope that qualified it, which excluded international orders under review. The post-compaction agent, reading its own record, escalates everything. No second system was ever involved.

WHAT EXISTING COVERAGE SHOWS – One agent, one healthy endpoint, no drift in the model, a memory system operating exactly as designed.

THE SLIP – A conversation between the agent and its former self, forked at the compression boundary. The writer no longer exists to be asked what the scope was. Only a record made at write time – of what the pre-compaction party understood the policy to be – makes the fork findable afterward.

# The Regulatory and Economic Landscape

## Deployer duties are arriving, written for single systems

Obligations to monitor, oversee, log, and report AI systems in consequential use are phasing in on fixed dates across jurisdictions, alongside the voluntary frameworks organizations are most often asked about.<sup>9,10</sup> Those duties were drafted with a single system in mind. As the systems they cover become conversational, “monitor the system” quietly comes to include the conversations inside it, and incident-reporting duties come to require an account of what the system’s parties understood. Organizations that can answer the four questions of the previous section will find those expectations descriptive rather than disruptive; organizations that kept no record of understanding will find some incident reports impossible to write.

## Counterparty conversations are third-party risk

Where agents transact across the organizational boundary, conversation governance meets an established discipline: third-party risk management. Today’s vendor questionnaires ask about the counterparty’s security, availability, and data handling. None yet asks who the counterparty’s agents are, what happens when their models change, or – when the two organizations’ agents disagree about what was agreed – whose record settles it. Contracts govern human communications exhaustively and are, at present, largely silent on what an agent’s utterance binds. These are questions organizations can begin asking of counterparties now, in the ordinary course of diligence, before the first disputed agent-formed commitment makes them urgent.

## Relation to intent monitoring

A body of recent work monitors agents for deviation from assigned intent – checking each action against the task before it executes,<sup>4</sup> or, in the most developed line, inferring an agent’s beliefs and goals from its full execution trace and classifying the trajectory as benign or malicious.<sup>5</sup> This work differs in both object and method. It makes no inference about intent: it measures displacement between parties’ understandings, and detection is the same whatever the cause.

The difference in coverage is structural. An intent monitor judges a trajectory against an authoritative statement of the task; this framework starts from the fact that no such record exists between conversing parties – each holds its own copy of what it believes is shared. And the benchmarks behind that literature define benign as the absence of a hidden objective,<sup>5</sup> so the common production case – two sincere parties, each correctly executing its own reading of an exchange they both believe they share, honestly wrong, together – has no hidden objective to find

and deviates from nothing either party was assigned. It passes intent monitoring by construction. The propensity battery described below produces exactly such traces, with planted ground truth, so the prediction is testable.

An intent-neutral tracker catches that case, and catches the rare sinister case the same way, as displacement in the record. The neutrality also protects the instrument itself. A monitor that assigns a cause at detection carries that reading forward – into the incident narrative, the strategies it learns, the next decision it makes – and if the reading was wrong, everything built on it is wrong the same way. A tracker that reads nothing into a slip has nothing to carry forward. It records what each party said it understood, when, and how far the accounts differ; interpretation comes later, from the record, and a wrong interpretation can be revised without the record inheriting it. Finally, where a trajectory monitor watches the boundary an organization assigns tasks across, the tracker covers the boundaries nothing assigns – between parties, and between a party and its former self, where the full trace an intent monitor would need has already been compacted away.

## The Measurement Program

This paper is the first artifact of a measurement program – the near-term work is as follows.

**A propensity battery.** A suite of controlled conversation starters targeting the first three signatures of the taxonomy – a planted false premise (absorbed premise), an ambiguous commitment (misassigned commitment), and a hedged fact relayed through a hop (hedge erosion) – run across pairs of current models, and scored not by judging transcripts but by asking each party, afterward and separately, what it understood, and comparing the answers against planted ground truth. The output is a propensity table: accommodation rates, commitment-assignment divergence, and hedge survival across hops, by model pair – including same-model pairs, where the results bear directly on the assumption that standardizing on one model closes the channel.

**A conversation tracker.** In the Language Model Tectonics™ family: an inventory of conversational pairs with both parties version-pinned; per-pair baselines established by a bounded calibration run and renewed when either party changes; periodic separate elicitation of each party's account of who owes what, with the accounts compared; and an append-only record from which every slip – induced, spontaneous, or hard – is reconstructible, so that when one matters, the timeline already exists. **Language Model Tectonics tells you when a model drifts. The slip tracker tells you when meaning does.**

**Facilitated assessment.** The published Language Model Diligence assessment covers a single model in a defined process. Conversational and agentic architectures – pipelines, orchestrations, systems with memory, and cross-boundary deployments – are assessed through facilitated engagement, using the four questions as pointed in this paper. Organizations operating such systems are encouraged to engage with Mossrake directly.

## Conclusion

Every organization deploying agentic AI should be able to answer four questions about the conversations its systems hold: who is talking to whom; whether anyone would know if the parties stopped meaning the same thing; whether understanding – not just output – is ever checked; and whether what each party understood could be reconstructed after the fact.

Most organizations, asked honestly, cannot yet answer any of them. That is not a failure of engineering or intent. The prompts were written carefully, and they govern what the agents say. Nothing yet governs what the agents understand – and between fluent parties, the difference does not announce itself. It accumulates quietly, releases suddenly, and its costs show up under other names.

The first response to this paper will be engineering, and it should be. Readback on consequential instructions, level-setting turns, summaries that preserve their hedges – these are real improvements, and deployments should make them. What no organization should permit itself is the sigh of relief that follows. An unbounded space does not become bounded because its most visible failures were addressed; there is always another layer, and the layers do not end. *The prompts are better* is this domain's *the tests pass* – true, valuable, and not an answer to the four questions above. The mature posture is the one every discipline in this paper's lineage arrived at: address the failure modes you can see, hedge the ones you cannot – or judge too small to chase – and keep the record that shows you knew the difference. That is not testing harder. It is risk management, and organizations that adopt it before the questions arrive from outside will find the eventual requirements descriptive rather than disruptive.

One limit belongs in plain sight. The point of tracking displacement as it accumulates is to intervene while it is small: an induced slip is caught with time to act, before anything is built on the fork, and that is how most harm from slip is stopped. What the watch cannot do is outrun an actor that moves faster than the checks. Against sustained misrepresentation the delay works in the record's favor – as between humans, the longer a party must keep a false account consistent with everything already on the record, the harder the lie gets. But for actions whose harm cannot be

repaired after the fact, the interval between checks is itself the exposure, and the right control there is a gate: the action does not execute until an independent party approves it. That is prevention, a different instrument, and high-consequence deployments should have it. The tracker is the layer that supports intervention – it surfaces divergence while there is still time to act on it, and it keeps the record by which the gates, the monitors, and the prompted repairs are themselves judged.

The question underneath is one commerce has asked for five hundred years: did the minds meet? What is new is not the question but the economics of asking it. No mind, human or machine, can be inspected; all anyone has ever had is conversation – the exchanges, the checkins, and a watch kept for slips. Humans keep that watch imperfectly, because every clarifying question costs something socially. A tracker outside the conversation pays no such cost. It can ask each party what it understands, while it understands it, on a schedule, and keep every answer. The watch that human interaction has always kept informally is, for these systems, an engineering decision away.

The first step is the same as it has always been in this discipline: know where you stand before anyone asks.

To take the Language Model Diligence self-assessment, or to arrange facilitated assessment of a conversational system, visit: [\*\*mossrake.ai/language-model-diligence\*\*](https://mossrake.ai/language-model-diligence)

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“Meeting of the minds” is used in this paper as working shorthand. The framework measures understanding between machine parties, not the formation or enforceability of contracts, and nothing in this paper is legal advice. Where agent-formed commitments raise legal questions, that is a matter for qualified counsel – who will find the conversation shorter if a record of what each party understood exists at all.

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